Problem Summary

Given the heads of two singly linked list, merge them into one sorted linked list. Return the head of the merged list.

Key Idea

Use a dummy node to build the new list.

Compare the front nodes of both lists:

→ attach the smaller one

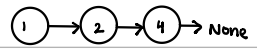
→ move that list forward

→ move curr forward

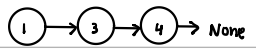
At the end, attach the remaining part

Diagram Example

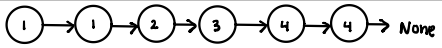
List 1:



List 2:



Result:



Approach

1) Create a dummy node

2) Let curr point to dummy

3) while both lists exist:

→ Compare list1.val and list2.val

→ attach smaller node to curr.next

→ move that list forward

→ move curr forward

4) Attach remaining nodes from unfinished list (if any)

5) Return dummy.next (head of resultant list)

Time Complexity: $O(n_1 + n_2)$ → since we visit atmost each node in each list

Space Complexity: $O(1)$ → only pointers are used